Thesis
Sophia Wang
October 22, 2026

---

1 Literature Review

Employer learning researches examine how firms update their assessments of workers as employment reveals information about productivity. Spence (1973) examines education as a way to signal ability when employers cannot observe true ability directly. Besides, Farber and Gibbons (1996) pay attention to wage dynamics, where measures of unobserved productivity should become more strongly associated with wages as experience accumulates. Further, Altonji and Pierret (2001) develop a related test using the NLSY79. Using wage regressions containing schooling and productivity proxy, they find that the association with the proxy strengthens while the schooling coefficient declines with experience. Though these papers differ in regression specifications and predictions for schooling, they share the same idea on information acquisition.

[Question on how to write a literature review]

2 Theoretical Model Setup

Let i index workers, $t \in \mathcal{T}_i$ be the observed year, k_{it} be the employer number, $j_{it} \in \{1, \ldots, J\}$ be the occupation, and $\tau < t$ represents earlier performance observations. Let e_{it} denote cumulative experience and x_{it} denote tenure with the current employer. Subscripts C and O represent the incumbent and the outside employer.

2.1 Two period Model

Altonji and Pierret (2001) specifies log productivity as a function of schooling s_i, employer observed information q_i,researcher observed ability estimation z_i, and remaining unobserved productivity η_i. Zhang instead starts from productivity in levels, $\chi_{if} = \eta_i + \delta_{if}$, where matching quality δ_{if} is known before production.

We formulate the equation as

$$p_{ijt}^k = a_j + r_j s_i + \alpha_{1j} q_i + \Lambda_j z_i + \eta_{ij} + \delta_i^k + H_j(e_{it}, x_{it})\,. \tag{1}$$

Here we assume p is the log productivity, so the productivity level is $Y = \exp(p)$. Following Zhang's assumption, the match component is know to the corresponding employer and is independently recalculated for new employers. The function $H_j(e, x)$ allows experience and tenure to affect productivity independently of employer learning.

Following Altonji and Pierret, assume

$$z_i = \gamma_1 q_i + \gamma_2 s_i + \nu_i \,, \quad \eta_{ij} = \alpha_{2j} s_i + e_{ij} \,, \tag{2}$$

where $\mathrm{E}[\nu_i \mid s_i, q_i] = 0$ and $\mathrm{E}[e_{ij} \mid s_i, q_i] = 0$.

Substituting eq. (2) into eq. (1) gives

$$\begin{aligned} p_{ijt}^k =& a_j + r_j s_i + \alpha_{1j} q_i + \Lambda_j \left(\gamma_1 q_i + \gamma_2 s_i + \nu_i\right) \\ &+ \alpha_{2j} s_i + e_{ij} + \delta_i^k + H_j(e_{it}, x_{it}) \,. \end{aligned}$$

Combining the predictable and initially unpredictable terms gives

$$p_{ijt}^k = B_{ij} + u_{ij} + \delta_i^k + H_j(e_{it}, x_{it}) \,, \tag{3}$$

where

$$B_{ij} = a_j + (r_j + \Lambda_j \gamma_2 + \alpha_{2j}) s_i + (\alpha_{1j} + \Lambda_j \gamma_1) q_i \,,$$

and

$$u_{ij} = \Lambda_j \nu_i + e_{ij} \,.$$

B_{ij} is the employer's initial assessment of log productivity in occupation j, and u_{ij} is its initial prediction error. Assume $u_{ij} \mid s_i, q_i \sim \mathcal{N}\left(0, \sigma_{0j}^2\right)$.

Altonji and Pierret construct a residual performance signal by subtracting employers' initial productivity assessment from observed noisy performance. Zhang's modification is that the performance signal is observed only by the incumbent.

During job k, let the employer observe

$$\xi_{i\tau}^k = p_{ij\tau}^k + \varepsilon_{i\tau}^k \,, \quad \varepsilon_{i\tau}^k \sim \mathcal{N}\left(0, \sigma_{\nu j}^2\right) \,,$$

with independent noise across observations and independent of persistent productivity.

Using eq. (3), we can subtract the known components

$$\begin{aligned} d_{i\tau}^k &= \xi_{i\tau}^k - B_{ij} - \delta_i^k - H_j(e_{i\tau}, x_{i\tau}) \\ &= u_{ij} + \varepsilon_{i\tau}^k \,, \end{aligned} \tag{4}$$

and define the employer's own performance record as

$$D_{it}^k = \left\{ d_{i\tau}^k : \tau < t, \tau \text{ belongs to job } k \right\} .$$

Deriving Zhang's learning weight

Suppose the first employer has observed n signals before the first mobility decision. From eq. (4),

$$\bar{d}_{it}^1 = u_{ij} + \bar{\varepsilon}_{it}^1 \,, \quad \mathrm{Var}(\bar{\varepsilon}_{it}^1) = \frac{\sigma_{\nu j}^2}{n} \,.$$

Therefore, $\mathrm{Cov}(u_{ij}, \bar{d}_{it}^1) = \sigma_{0j}^2$, and $\mathrm{Var}(\bar{d}_{it}^1) = \sigma_{0j}^2 + \frac{\sigma_{\nu j}^2}{n}$. As these variables are jointly normal, the conditional mean is

$$\begin{aligned} \mathrm{E}[u_{ij} \mid s_i, q_i, D_{it}^1] &= \frac{\mathrm{Cov}(u_{ij}, \bar{d}_{it}^1)}{\mathrm{Var}(\bar{d}_{it}^1)} \, \bar{d}_{it}^1 \\ &= \frac{\sigma_{0j}^2}{\sigma_{0j}^2 + \sigma_{\nu j}^2/n} \, \bar{d}_{it}^1 \,. \end{aligned}$$

Hence,

$$\hat{u}_{ijt}^{C,1} = \theta_j(n) \bar{d}_{it}^1, \quad \theta_j(n) = \frac{n\sigma_{0j}^2}{\sigma_{\nu j}^2 + n\sigma_{0j}^2} \,. \tag{5}$$

The corresponding posterior variance is

$$V_j(n) = \left(\frac{1}{\sigma_{0j}^2} + \frac{n}{\sigma_{\nu j}^2} \right)^{-1} = \frac{\sigma_{0j}^2 \cdot \sigma_{\nu j}^2}{\sigma_{\nu j}^2 + n\sigma_{0j}^2} \,,$$

where n represents the performance observations, which equals tenure measured in each job periods. Specifically, for $n = 1$, eq. (5) matches with Zhang's eq. (1).

Taking expectations in eq. (3) and then using eq. (5),

$$\mathrm{E}[p_{ijt}^1 \mid \mathcal{I}_{it}^C] = B_{ij} + \delta_i^1 + H_j(e_{it}, x_{it}) + \theta_j(n) \bar{d}_{it}^1 \,. \tag{6}$$

Substitute

$$\bar{d}^1_{it} = \Lambda_j(z_i - \gamma_1 q_i - \gamma_2 s_i) + e_{ij} + \bar{\varepsilon}^1_{it}$$

into eq. (6), we get

$$\begin{aligned}\mathrm{E}[p^1_{ijt} \mid \mathcal{I}^C_{it}] =& a_j + [r_j + \alpha_{2j} + (1 - \theta_j(n))\Lambda_j\gamma_2]\, s_i \\ &+ [\alpha_{1j} + (1 - \theta_j(n))\Lambda_j\gamma_1]\, q_i \\ &+ \theta_j(n)\Lambda_j z_i + \theta_j(n)e_{ij} + \theta_j(n)\bar{\varepsilon}^1_{it} + \delta^1_i + H_j(e_{it}, x_{it})\,.\end{aligned}$$

For $F \in \{C, O\}$, define

$$P^F_{it} = \mathrm{E}[Y^F_{it} \mid \mathcal{I}^F_{it}] = \exp[R_{ij^F} + \delta^F_i + H_{j^F}(e_{it}, x^F_{it})]\mathrm{E}[\exp(u_{ij^F}) \mid \mathcal{I}^F_{it}]\,, \tag{7}$$

where j^C is the incumbent occupation, j^O the offered occupation, and $x^O_{it} = 0$ for a new match. For the first incumbent's normal posterior, eq. (5) and eq. (7) imply x

$$P^O_{it_1} = \exp\left[B_{ij^O} + \delta^O_i + H_{j^O}(e_{it_1}, 0) + \frac{1}{2}\sigma^2_{0j^O}\right]\,.$$

Endogenous mobility and the informational content of employment history

Zhang assumes that the outsider offers first, then the incumbent counteroffers, and finally the worker compares wages plus non-pecuniary job preferences. Let

$$\theta^C_{it}, \theta^O_{it} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2_\theta)\,, \quad s_\theta = \sqrt{2}\sigma_\theta\,.$$

The worker moves when

$$W^O_{it} + \theta^O_{it} \geq W^C_{it} + \theta^C_{it}\,.$$

Since the taste difference is $\mathcal{N}(0, 2\sigma^2_\theta)$,

$$\Pr(M_{it} = 1 \mid W^C_{it}, W^O_{it}) = \Phi(A_{it})\,, \quad A_{it} = \frac{W^O_{it} - W^C_{it}}{s_\theta}\,. \tag{8}$$

From $\chi_c = \eta + \delta_c$ where $n \sim \mathcal{N}(\mu_n, \sigma_\eta^2)$, and $y_{c1} = \chi_c + \varepsilon_{c1}$ where $\varepsilon_{c1} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, define

$$\begin{aligned} P^C &\equiv \mathrm{E}[\chi_c \mid y_{c1}, \delta_c] = \mu_n + \delta_c + \frac{\sigma_\eta^2}{\sigma_\eta^2 + \sigma_\varepsilon^2}[y_{c1} - (\mu_\eta + \delta_c)] \\ &= \left(1 - \frac{\sigma_\eta^2}{\sigma_\eta^2 + \sigma_\varepsilon^2}\right)(\mu_\eta + \delta_c) + \frac{\sigma_\eta^2}{\sigma_\eta^2 + \sigma_\varepsilon^2}\, y_{c1} \\ &= \frac{\sigma_\varepsilon^2}{\sigma_\eta^2 + \sigma_\varepsilon^2}(\mu_\eta + \delta_c) + \frac{\sigma_\eta^2}{\sigma_\eta^2 + \sigma_\varepsilon^2}\, y_{c1}\,. \end{aligned}$$

As the outside employer does not observe y_{c1}, in the two period specification, its assessment will rely on the population mean of ability and its own known match quality. Therefore, we can write

$$P^O \equiv \mathrm{E}[\eta + \delta_o \mid \delta_o] = \mu_\eta + \delta_o\,.$$

Then following Zhang's eq. (2) and (3), we can construct optimization problem in the format of

$$\max_{W^C}\left(P^C - W^C\right)(1 - \Phi(A)) \quad \text{and} \quad \max_{W^O}(P^O - W^O)\Phi(A)\,.$$

Taking first order derivative to W^C and W^O, respectively, gives

$$\begin{cases} W^C = & P^C - s_\theta \frac{(1-\Phi(A))}{\phi(A)}\,, \\ W^O = & P^O - s_\theta \frac{\Phi(A)}{\phi(A)}\,. \end{cases}$$

As $((1 - \Phi(A))/\phi(A))' < 0$, holding the outside offer fixed, differentiate W^C equation above gives

$$dW^C = dP^C - s_\theta \left(\frac{1 - \Phi(A)}{\phi(A)}\right)' dA\,. \tag{9}$$

Holding the outside offer W^O and s_θ fixed,

$$dA = -\frac{dW^C}{s_\theta}\,.$$

Substituting this into (9) gives

$$dW^C = dP^C - s_\theta \left(\frac{1 - \Phi(A)}{\phi(A)}\right)' \left(-\frac{dW^C}{s_\theta}\right) = dP^C + \left(\frac{1 - \Phi(A)}{\phi(A)}\right)' dW^C\,.$$

Using the fact that $\Phi'(A) = \phi(A)$ and $\phi(A) = 1/\sqrt{2\pi} \cdot e^{-A^2/2} \Rightarrow \phi'(A) = -A\phi(A)$,

$$\left.\frac{\partial W^C}{\partial P^C}\right|_{W^O, s_\theta} = \frac{1}{1 - \left(\frac{1-\Phi(A)}{\phi(A)}\right)'} = \frac{1}{2 - A\left(\frac{1-\Phi(A)}{\phi(A)}\right)}. \tag{10}$$

As

$$\left(\frac{1-\Phi(A)}{\phi(A)}\right)' = -1 + A\left(\frac{1-\Phi(A)}{\phi(A)}\right) < 0\,,$$

We have

$$0 < \frac{\partial W^C}{\partial P^C} < 1\,,$$

which indicates that holding outside offer and taste dispersion fixed, an increase in the incumbent's assessment of worker productivity raises its wage offer by less than the productivity increase. The positive response reflects the employer's stronger incentive to

an increase in the incumbent's expected productivity raises its wage offer, but retain a worker it now considers more productive. However, the worker receives only part of the additional expected productivity through higher wages; the remainder increases the employer's marginal profit conditional on retention. Thus, favorable information acquired through employer learning raises wages, but its effect is only partially transmitted to the worker. The implications for wage growth with tenure or experience depend on how these variables affect productivity assessments and outside offers.

3 Data, Sample Construction, and Measurement

3.1 Datasets

{Follow Altonji's data structure, e.g. add description of dataset scope}

The empirical analysis uses the National Longitudinal Survey of Youth 1979 (NLSY79) and the National Longitudinal Survey of Youth 1997 (NLSY97). Both surveys provide longitudinal information on wages, employment histories, occupations, education, demographics, and Armed Services Vocational Aptitude Battery (ASVAB) test scores, allowing respondents to be followed across employers and over labor-market experience. Due to difference in the employment history structure, the two datasets are cleaned separately. In both datasets, negative survey codes are treated as missing values. Hourly wages have been converted from cents to dollars and adjusted to 2023 real dollars using the CPI.

For the NLSY79 dataset, employment durations, which are measured in years, are ordered chronologically within each job period of the worker and are accumulated to construct actual labor market experience, with having the reported within job gaps subtracted from employment durations. {Conclusion comes first, then explanations} For each of the worker's job, education is fixed at the value observed at the commencement of this job. Therefore. schooling does not change with subsequent labor market tenure and experience increase. For the NLSY97 dataset, employers are identified using employer Unique ID (UID). The main employer {Ask Smith if this is standard in literature} is first identified in each survey year. When it changes between adjacent surveys, the employment roster is used to distinguish the previous and new employers {Explain in footnote}. Employer spells are then numbered sequentially within one worker according to the sequence of employer UID. The main analysis will focus on workers' first three jobs, while workers are not required to be observed in all three jobs.

3.2 Dependent and Independent Variables

[Adding citations of reasons for using asvab not afqt]

The dependent variable is the log real hourly wage, denote by $\ln w_{it}$, where w_{it} is the CPI adjusted hourly wage of worker i in year t.

The main independent variables are schooling, ability, labor-market experience, employer tenure, occupation, and prior mobility. Schooling is measured using education observed at the beginning of the first employer spell and is then held fixed over the worker's subsequent employment history.

Worker ability is constructed from ASVAB subtest scores. In the NLSY79, individual ASVAB components are standardized within birth-year-by-quarter groups, after which principal-components analysis is applied. The first principal component is used as the ability measure and explains approximately 68.6% of the variation in the standardized scores.The NLSY97 ASVAB variables are first reconstructed by combining the positive and negative ability estimates for eight subtests: General Science, Arithmetic Reasoning, Word Knowledge, Paragraph Comprehension, Numerical Operations, Coding Speed, Mathematics Knowledge, and Mechanical Comprehension. The subtests are then standardized using the same birth-year-by-quarter procedure and summarized using the first principal component, which explains approximately 67.2% of their standardized variation.

For both cohorts, actual experience is calculated from cumulative weeks worked after subtracting within-job gaps. Employer tenure measures duration with the current employer.

For both cohorts, employer spells are ordered chronologically, and prior mobility can be summarized by the number of previous employer changes.

Following grouping method used by Gibbons et al. (2005), we mapped the occupation codes into seven major groups: Professional, Managers, Sales, Clerical, Craftsmen, Operatives and Laborers, and Service, under the 1970 census occupation grouping method. For observations reported using the 2000/2002 Census occupation classification, we examine the occupational definitions of each code and assign them to the 1970 group that most closely corresponds to its occupational content.

3.3 Summary Statistics

Table 1
Summary Statistics

Variable	Observations	Mean	SD	Minimum	Maximum
Panel A: Characteristics across observations					
Experience	55,972	5.030	5.319	0	44.327
Hours	55,972	34.661	12.937	0	168
Tenure	55,972	3.427	4.433	0	29.635
Wage	55,972	18.289	11.914	2.832	110.829
Cohort indicator	55,972	0.204	0.403	0	1
Age	55,972	24.447	5.848	15	47
Panel B: Individual characteristics					
ASVAB score	13,154	0.005	2.341	−7.300	7.324
Education	13,154	11.627	1.811	0	20
Sex category	13,154	1.491	0.500	1	2

Panel C: Occupational composition (%)	NLSY79	NLSY97	Pooled
Clerical	22.78	25.39	23.31
Craftsmen	9.58	7.47	9.15
Managers	6.47	5.65	6.30
Operatives and laborers	24.49	16.02	22.76
Professional	8.68	16.64	10.31
Sales	4.76	6.70	5.16
Service	23.25	22.12	23.02
Total	100.00	100.00	100.00

Panel A reports statistics across observations. Panel B uses one record per respondent, with education averaged across that respondent's observations. Sex category equals 1 for female and 2 for male. The cohort indicator equals 0 for NLSY79 and 1 for NLSY97. Panel C reports occupation percentages within each cohort and in the pooled sample.

4 Empirical Specifications

4.1 OLS Model

{Tell conclusion at the beginning of this section. e.g. in this section we will show that... to show this, we begin by ... }

{First we use this model... later we introduce a more complete one..}

We begin by examining how the relationship among wage, schooling, and ASVAB evolves over workers' first three job spells. In this section, we will show that wage becomes more closely associated with ability as employers learn information about productivity. {and whether this relationship varies with employment history.} {Change sentence to recall}We distinguish cumulative labor market experience from tenure with the current employer and estimate the specification separately from the NLSY79 and NLSY97 cohorts.

{Mention these three articles's conclusions} The specification comes from three articles. Altonji and Pierret (2001) motivate examining whether wages become more closely associated with initially unobserved ability as employers learn, while reliance on schooling as a productivity signal declines. Gibbons et al. (2005) emphasize allowing wage coefficients to vary across occupations, reflecting differences in the valuation of worker skills. Zhang (2009) distinguishes cumulative experience from employer tenure, where incumbent employers observe performance within the employment, whereas outside employers can infer productivity from employment histories. The resulting equation is a empirical specifications constructed by these mechanisms.

We first estimate a baseline equation separately for each cohort, without allowing wage relationships to vary across occupations. We will discuss occupation variation later in this section. Following the employer learning motivation of Altonji and Pierret (2001), we allow the associations of schooling and measured ability with wages to change with experience. We extend this specification by including interactions with employer tenure:

$$\begin{aligned}\log w_{itc} =& \alpha_{kc} + \beta_{Sc} S_{ikc} + \beta_{Zc} Z_{ic} + \gamma_{Sc}\left(S_{ikc} E_{itc}\right) + \gamma_{Zc}\left(Z_{ic} E_{itc}\right) \\ &+ \theta_{Sc}\left(S_{ikc} T_{itc}\right) + \theta_{Zc}\left(Z_{ic} T_{itc}\right) + f_c\left(E_{itc}\right) + g_c\left(T_{itc}\right) \\ &+ \lambda_{tc} + X_{itc}\eta_c + \varepsilon_{itc}\,. \end{aligned} \tag{11}$$

Let i be the worker index, t represent calendar years, and $c \in \{79, 97\}$ indicate survey cohorts. Let $k = k_{it} \in \{1, 2, 3\}$ represent worker's current job spell, $j = j_{it} \in \{1, \ldots, J\}$ represent the current occupation, and D_{ijtc} indicate the employment in occupation j.

Here, w_{itc} is the real hourly wage, S_{ikc} is schooling measured at the beginning of spell $k = 1$ and held fixed within that job spell, and Z_{ic} is the standardized ASVAB based ability measure, which is observed by the researcher but assumed not to be directly observed by employers. Actual experience, E_{itc}, accumulates across jobs, and tenure, T_{itc}, measures employment with the current employer, where both of these two variables are measured in years. Occupation indicators refer to the occupation reported at survey time each year.

The functions f_c and g_c are cubic polynomials in the form of $x + x^2 + x^3$. The function h_{jc} contains occupation specific deviations in the same experience and tenure polynomial terms. Therefore, occupational differences in ordinary wage growth are accommodated alongside differences in the schooling and ability gradients. The specification includes job spell effects α_{kc}, calendar year effects λ_{tc}, additional demographical controls X_{itc}, and standard errors ε_{itc} which are clustered by worker.

The implied schooling and ability wage gradients are

$$
\begin{aligned}
b_{Sc}(E,T) &= \beta_{Sc} + \gamma_{Sc}E + \theta_{Sc}T\,, \\
b_{Zc}(E,T) &= \beta_{Zc} + \gamma_{Zc}E + \theta_{Zc}T\,.
\end{aligned}
$$

Under the informational assumptions of the employer learning model, an increasing ability gradient and a decreasing schooling gradient are consistent with employers placing greater weight on observed productivity and less weight on schooling as a signal. At fixed tenure, these patterns correspond to $\gamma_{Zc} > 0$ and $\gamma_{Sc} < 0$. During the continuous employment with the same employer, experience and tenure both increase. The corresponding changes in the gradients are therefore $\gamma_{Zc} + \theta_{Zc}$ and $\gamma_{Sc} + \theta_{Sc}$. These estimates provide descriptive evidence on employer learning, as skill accumulation and worker sorting can also change wage gradients.

Including occupational categories, we then formulate a more complete regression model

$$
\begin{aligned}
\log w_{itc} =& \alpha_{kc} + \beta_{Sc}S_{ikc} + \beta_{Zc}Z_{ic} + \gamma_{Sc}S_{ikc}E_{itc} + \gamma_{Zc}Z_{ic}E_{itc} \\
&+ \theta_{Sc}S_{ikc}T_{itc} + \theta_{Zc}Z_{ic}T_{itc} + f_c\left(E_{itc}\right) + g_c\left(T_{itc}\right) \\
&+ \sum_{j=2}^{J} D_{ijtc}\big[\delta_{jc} + \rho_{Sjc}S_{ikc} + \rho_{Zjc}Z_{ic} + h_{jc}\left(E_{itc}, T_{itc}\right) + \phi_{Sjc}\left(S_{ikc}E_{itc}\right) \\
&+ \phi_{Zjc}\left(Z_{ic}E_{itc}\right) + \psi_{Sjc}\left(S_{ikc}T_{itc}\right) + \psi_{Zjc}\left(Z_{ic}T_{itc}\right)\big] + \lambda_{tc} + X_{itc}\delta_c + \varepsilon_{itc}\,. \qquad (12)
\end{aligned}
$$

{Change this sentence to explain occ 1} Occupation 1 is the reference category. For $q \in$

$\{S, Z\}$, the conditional wage coefficient in non-reference occupations ($j \neq 1$) is

$$b_{qjc}(E,T) = \beta_{qc} + \rho_{qjc} + (\gamma_{qc} + \phi_{qjc})\, E + (\theta_{qc} + \psi_{qjc})\, T \tag{13}$$

{add explain of occupation deviation} with the occupational deviations set to zero for $j = 1$. Consequently, ϕ_{qjc} and ψ_{qjc} measure occupational differences in how the schooling β_{Sc} or ability β_{Zc} coefficient varies with experience and tenure, respectively.

During the continued employment within a given occupation, both of experience and tenure increase. Therefore, we define the changes in the wage ability gradient as

$$L_{Zjc} \equiv \gamma_{Zc} + \phi_{Zjc} + \theta_{Zc} + \psi_{Zjc}\,, \tag{14}$$

$$L_{Sjc} \equiv \gamma_{Sc} + \phi_{Sjc} + \theta_{Sc} + \psi_{Sjc}\,. \tag{15}$$

Employer learning predicts $L_{Zjc} > 0${footnote explain why Lzjc>0}. If schooling initially serves as a productivity signal, the analogous schooling gradient is expected to decline, therefore $L_{Sjc} < 0$ {footnote explain again}. Comparisons of these quantities across occupations and cohorts supports the principal empirical questions' opinion: {mention empirical question again}. When experience and tenure coincide in the first spell, separating their contributions requires variation from subsequent spells.

{explicitly state result change in this paragraph}The baseline OLS estimates allow us to compare how the wage associations of schooling and ability change with experience and tenure across occupations and cohorts. Results consistent with the predicted signs provide suggestive evidence of employer learning. {Specific}However, these patterns may also reflect differences in skill development, the rewards to skills, and workers' selection into jobs. The estimates therefore cannot establish whether occupational or cohort differences arise from employer learning alone.

4.2 Instrument Selection

As shown in 2, in both of the two regressions, cubic experience and cubic tenure yield numerically small coefficients. For consideration of simplicity, we in this section adapt a quadratic experience and tenure specification. This reduces the number of endogenous terms while still capturing experience profile curvature. Let E_{it} denote actual experience, T_{it} employer tenure, S_{it} schooling, A_i the ASVAB score, D_{jit} the occupation j indicator,

and X_{it} demographic and year controls. The new proposed wage equation is

$$\begin{aligned}100 \ln w_{it} =& \alpha + \beta_S S_{it} + \beta_A A_i + \gamma_1 E_{it} + \gamma_2 E_{it}^2 + \tau_1 T_{it} + \tau_2 T_{it}^2 \\ &+ \lambda_S S_{it} E_{it} + \lambda_A A_i E_{it} + \theta_S S_{it} T_{it} + \theta_A A_i T_{it} + \sum_{j=2}^{7} D_{jit} [\mu_j + \beta_{Sj} S_{it} + \beta_{Aj} A_i \\ &+ \lambda_{Sj} S_{it} E_{it} + \lambda_{Aj} A_i E_{it} + \theta_{Sj} S_{it} T_{it} + \theta_{Aj} A_i T_{it}] + X_{it}' \delta + u_{it} \,.\end{aligned}$$

Actual experience and occupational may both due to productivity information unobserved by the researcher. In this section, we will therefore treat experience, occupation, and their interactions with other variables as endogenous. The endogenous regressor vector is

$$E_{it}, E_{it}^2, S_{it} E_{it}, A_i E_{it}, \{D_{jit}, D_{jit} S_{it}, D_{jit} A_i, D_{jit} S_{it} E_{it}, D_{jit} A_i E_{it}\}_{j=2}^{7}$$

$$V_{it} = \left(\begin{array}{c} E_{it}, E_{it}^2, T_{it}, T_{it}^2, S_{it} E_{it}, A_i E_{it}, S_{it} T_{it}, A_i T_{it}, \\ \{D_{jit}, D_{jit} S_{it}, D_{jit} A_i, D_{jit} S_{it} E_{it}, D_{jit} A_i E_{it}, D_{jit} S_{it} T_{it}, D_{jit} A_i T_{it}\}_{j=2}^{7} \end{array} \right)^T .$$

Let V_{kit} denote the $k$the element in V_{it}, Z_{it} represent the excluded instruments, and the exogenous regressor as

$$W_{it} = (1, S_{it}, A_i, X_{it}')' \,.$$

For each endogenous term V_{kit}, the first stage equation is

$$V_{kit} = Z_{it}' \pi_k + W_{it}' \rho_k + \eta_{kit} \,, \quad k = 1, \ldots K \,.$$

where $K = \dim(V_{it})$ and η_{kit} is the first stage error term. Estimate each equation by OLS yields

$$\hat{V}_{kit} = Z_{it}' \hat{\pi}_j + W_{it}' \hat{\rho}_k \,.$$

Actual labor market experience may be endogenous because employment continuity depends on worker productivity, labor supply decisions, or employer assessments. Occupational assignment and job duration may similarly respond to information about productivity. Therefore, instrument variables are helpful to distinguish endogeneity of regressors.

Altonji and Pierret (2001) instrument actual experience terms with corresponding functions of potential experience, due to the fact that realized employment may itself convey information about worker's productivity. Zhang (2009)follows this approach in his baseline and subsequently distinguishes experience from tenure when examining employer

learning. Gibbons et al. (2005) address endogenous occupational sorting using instruments constructed from sector histories.

Since actual experiences are included in the model through polynomial terms and interactions with schooling, ability, and occupation. Each of these terms is treated as a separate endogenous regressor. Identifying their coefficients therefore requires at least as many linearly independent excluded instruments as endogenous regressors, together with a full-rank relationship between the instruments and these regressors conditional on the included exogenous variables.

Let P_{itc} denote potential experience, measured as elapsed years since the worker's initial job market entry. This measure accumulates across the first three job spells and includes periods without employment. We adapt the potential experience approach by replacing each actual experience term with its corresponding function of P_{itc}. For the common experience terms, the excluded instrument vector is

$$\left(P_{itc}, P_{itc}^2, P_{itc}^3, S_{ikc}P_{itc}, Z_{ic}P_{itc}\right) .$$

To accommodate occupational differences, we additionally interact these instruments with the non-reference occupational indicators, where the vector of instruments is

$$\left(Q_{itc}^0, D_{i2tc}Q_{itc}^0, \ldots D_{iJtc}Q_{itc}^0\right) .$$

This construction matched the polynomial order and occupational interactions in the baseline equation. Therefore we have $5 \cdot J$ excluded instruments for the $5 \cdot J$ experience related regressors in the baseline equation. Restricting the instruments to the corresponding polynomial terms and occupational interactions limits their number and the potential for first-stage overfitting.

5 Results

5.1 Baseline Model Analysis

In this subsection, we will examine how wage associates with schooling and measured ability evolve with experience and employer tenure in the NLSY79 and NLSY97 cohorts. Table 2 presents the baseline estimates with common coefficients across occupations, while Table 3 allows the coefficients to vary across occupational groups. In the analysis, we

will compare coefficient magnitudes and statistical significance across cohorts, distinguish the schooling and ability gradients at zero experience and tenure from their subsequent changes. Linear combination tests assess whether these gradients change during continued employment with the same employer, while joint Wald tests assess whether the corresponding coefficients differ across occupations. We then compare the findings with the employer learning evidence of Altonji and Pierret (2001) and the analysis of skill valuation and occupational sorting in Gibbons et al. (2005), evaluating which patterns are consistent with employer learning and which will require further explanations.

5.2 IV Model

Baseline

The baseline IV model without occupation terms relates log wages to schooling S, ASVAB Z, cubic experience and tenure terms, and schooling and ASVAB interactions with linear experience and tenure, controlling for job spell, year, sex, and race. We instrument (E, E^2, E^3, SE, ZE) with (P, P^2, P^3, SP, ZP), where P measures years since first observed employment. Other regressors are treated as exogenous.

Table 2
Education, Ability, and Wage Dynamics

	Baseline		Occupation model	
	NLSY79	NLSY97	NLSY79	NLSY97
	(1)	(2)	(3)	(4)
A. Regression coefficients				
Education	7.291^{***}	3.508^{***}	4.533^{***}	3.680^{**}
	(0.349)	(0.960)	(0.540)	(1.517)
ASVAB	-1.320^{***}	0.344	-1.258^{***}	0.200
	(0.266)	(0.714)	(0.393)	(0.865)
Experience	6.563^{***}	-0.273	4.130^{*}	13.178^{**}
	(1.379)	(2.723)	(2.336)	(5.673)
p value	<0.0001	0.9203	0.0771	0.0202
Experience^2	-0.244^{***}	-0.137	-0.143^{**}	-0.547^{***}
	(0.034)	(0.113)	(0.068)	(0.176)
p value	<0.0001	0.2237	0.0356	0.0019
Experience^3	0.00275^{***}	0.00065	0.00114	0.01267^{**}
	(0.00058)	(0.00343)	(0.00138)	(0.00535)
p value	<0.0001	0.8501	0.4088	0.0179
Tenure	9.094^{***}	4.788^{*}	9.088^{***}	-1.862
	(1.828)	(2.689)	(3.290)	(4.953)
p value	<0.0001	0.0751	0.0057	0.7069
Tenure^2	-0.728^{***}	-0.265^{*}	-0.706^{***}	0.008
	(0.059)	(0.136)	(0.093)	(0.216)
p value	<0.0001	0.0510	<0.0001	0.9717
Tenure^3	0.01628^{***}	0.00623	0.01616^{***}	-0.00505
	(0.00180)	(0.00474)	(0.00278)	(0.00688)
p value	<0.0001	0.1884	<0.0001	0.4625
Education × Experience	-0.101	0.316	0.002	-0.474
	(0.107)	(0.202)	(0.189)	(0.441)
ASVAB × Experience	0.128^{*}	0.132	0.291^{*}	0.365
	(0.072)	(0.190)	(0.163)	(0.225)
Education × Tenure	0.141	0.036	0.042	0.461
	(0.139)	(0.202)	(0.270)	(0.381)
ASVAB × Tenure	0.364^{***}	0.100	0.064	-0.329
	(0.101)	(0.215)	(0.233)	(0.298)
B. Linear combination tests				
Education, $\gamma_S + \theta_S$	0.040	0.353^{**}	0.044	-0.012
	(0.086)	(0.174)	(0.189)	(0.287)
p value	0.6377	0.0429	0.8166	0.9653
ASVAB, $\gamma_Z + \theta_Z$	0.492^{***}	0.232^{*}	0.355^{**}	0.036
	(0.070)	(0.129)	(0.141)	(0.215)
p value	<0.0001	0.0718	0.0119	0.8686
C. Tests of cubic terms: p values				
Experience cubic terms = 0	<0.0001	0.8501	<0.0001	0.0826
Tenure cubic terms = 0	<0.0001	0.1884	<0.0001	0.0113
All cubic terms = 0	<0.0001	0.2195	<0.0001	0.0022
Year and job number indicators	Yes	Yes	Yes	Yes
Sex and race controls	Yes	Yes	Yes	Yes
Occupation indicators and interactions	No	No	Yes	Yes
Observations	57,214	11,440	57,214	11,440
Respondents	11,065	2,889	11,065	2,889
R^2	0.3724	0.2933	0.4079	0.3455

The outcome is 100 times log real hourly wages. Regressions use sampling weights; standard errors clustered by respondent are in parentheses. Columns (3) and (4) report clerical coefficients and linear combinations, while other occupation groups are not displayed. Panel C tests the common cubic coefficients for baseline models and all occupation cubic coefficients for occupation models.
$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 3
Education, Ability, and Wage Dynamics by Occupation

Occupation	Education			ASVAB		
	Level (1)	× Experience (2)	× Tenure (3)	Level (4)	× Experience (5)	× Tenure (6)
A. NLSY79						
Clerical	4.690^{***}	-0.111	0.112	-1.032^{**}	0.330^{*}	-0.070
	(0.588)	(0.204)	(0.206)	(0.423)	(0.178)	(0.200)
Craftsmen	4.908^{***}	0.325	-0.574	0.051	-0.070	0.379^{**}
	(1.499)	(0.272)	(0.414)	(0.962)	(0.142)	(0.178)
Managers	5.800^{***}	-0.317	0.575	0.853	0.503	-0.422
	(2.049)	(0.325)	(0.350)	(1.399)	(0.336)	(0.356)
Operatives and	5.674^{***}	-0.178^{*}	0.506^{**}	-0.730	0.003	0.230
laborers	(0.515)	(0.107)	(0.211)	(0.461)	(0.088)	(0.189)
Professional	7.662^{***}	0.362	-0.584^{**}	1.550	-0.175	0.442^{**}
	(1.072)	(0.240)	(0.243)	(1.078)	(0.202)	(0.216)
Sales	6.503^{***}	-0.806^{***}	1.671^{***}	-2.583^{***}	0.684^{***}	0.074
	(1.374)	(0.245)	(0.523)	(0.965)	(0.197)	(0.432)
Service	7.227^{***}	-0.202	0.113	-0.893^{**}	0.075	0.364^{*}
	(0.518)	(0.147)	(0.170)	(0.352)	(0.096)	(0.190)
Equality test, p	0.0235	0.0228	<0.001	0.0717	0.0153	0.2179
Observations: 59,927; Respondents: 11,072; $R^2 = 0.4410$.						
B. NLSY97						
Clerical	3.680^{**}	-0.474	0.461	0.200	0.365	-0.329
	(1.517)	(0.441)	(0.381)	(0.865)	(0.225)	(0.298)
Craftsmen	2.608	0.418	-0.351	-3.596	-0.005	0.120
	(4.758)	(0.864)	(0.844)	(3.732)	(0.829)	(0.752)
Managers	12.092^{***}	0.670	-1.013	0.008	-0.295	0.882
	(2.672)	(0.626)	(0.643)	(3.301)	(0.674)	(0.749)
Operatives and	1.121	-0.121	0.371	0.820	0.224	-0.338
laborers	(1.417)	(0.258)	(0.298)	(1.216)	(0.312)	(0.352)
Professional	3.559^{**}	0.788^{**}	-0.383	-2.081	1.013^{**}	-0.680
	(1.678)	(0.340)	(0.374)	(1.767)	(0.446)	(0.529)
Sales	9.793^{*}	-0.064	-0.184	-1.227	-0.454	0.758
	(5.422)	(0.979)	(1.233)	(1.851)	(0.556)	(0.784)
Service	0.856	0.313	0.150	0.920	0.722^{***}	-0.518
	(1.382)	(0.316)	(0.400)	(1.010)	(0.274)	(0.386)
Equality test, p	0.0041	0.2295	0.2977	0.6619	0.2827	0.4784
Observations: 11,440; Respondents: 2,889; $R^2 = 0.3455$.						

Estimates and standard errors are expressed on the scale of 100 times log real hourly wages. Using sampling weights, and standard errors clustered by respondent are in parentheses. Wages are winsorized at the 1st and 99th percentiles of the pooled merged dataset before restricting job numbers to 1 through 3. Education, experience, and tenure are in years. ASVAB is in principal component units. Level coefficients apply at zero experience and tenure.

$^{*}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Table 4
IV models: First stage relevance

	NLSY79		NLSY97	
Endogenous regressor	Robust F	Shea R^2	Robust F	Shea R^2
Panel A. Baseline model				
E	81.60	0.263	138.85	0.395
E^2	29.87	0.094	61.84	0.322
E^3	17.90	0.056	39.18	0.326
SE	81.05	0.389	146.36	0.554
ZE	39.72	0.360	47.24	0.517
Observations	30,246		11,438	
Respondents	7,631		2,889	
Panel B. Model with occupation indicators				
E	17.91	0.367	28.89	0.459
E^2	9.17	0.116	19.41	0.383
E^3	5.62	0.069	17.10	0.375
SE	18.75	0.499	31.91	0.631
ZE	12.73	0.425	29.99	0.673
Professional	59.56	0.386	142.63	0.693
Managers	33.30	0.309	1,354.73	0.633
Sales	11.09	0.213	5,401.60	0.688
Craftsmen	35.37	0.350	3,610.77	0.772
Operatives and laborers	55.77	0.362	371.14	0.712
Service	78.47	0.408	317.92	0.713
Observations	15,089		4,555	
Respondents	3,746		1,111	
Excluded instruments	53		52	

Notes. E is actual experience, S schooling, Z ASVAB, and P years since first observed employment. Panel A instruments (E, E^2, E^3, SE, ZE) with (P, P^2, P^3, SP, ZP). Panel B also treats six occupation indicators as endogenous; clerical is the reference category. Panel B adds indicators for occupations in the previous two survey waves and their interactions to the excluded instruments.

Table 5
Baseline IV: First and second stage estimates

	NLSY79		NLSY97	
Variable Dependent variable	First stage Experience	Second stage $100\ln(wage)$	First stage Experience	Second stage $100\ln(wage)$
Schooling	-0.0056	7.5906^{***}	0.0685	4.1459^{***}
	(0.0352)	(0.4549)	(0.0502)	(1.1067)
Schooling × experience		-0.0588		0.0935
		(0.2502)		(0.2858)
Schooling × tenure	-0.0202	-0.1352	-0.0178	0.2384
	(0.0171)	(0.2783)	(0.0124)	(0.2663)
ASVAB	-0.0626^{**}	-1.7305^{***}	-0.0679^{*}	-0.2306
	(0.0301)	(0.3443)	(0.0390)	(0.7928)
ASVAB × experience		0.5858^{**}		0.3683
		(0.2352)		(0.2840)
ASVAB × tenure	-0.0520^{***}	0.0226	-0.0133	-0.1312
	(0.0153)	(0.2551)	(0.0122)	(0.2968)
Experience		3.5307		1.6613
		(3.8034)		(4.1191)
Experience^2		-0.1728		0.0371
		(0.1790)		(0.1999)
Experience^3		0.0048		-0.0018
		(0.0048)		(0.0059)
Tenure	1.0772^{***}	13.9425^{***}	0.8375^{***}	2.4244
	(0.2047)	(3.9642)	(0.1642)	(3.7679)
Tenure^2	-0.0384^{***}	-0.8908^{***}	-0.0228^{***}	-0.3831^{**}
	(0.0058)	(0.1566)	(0.0061)	(0.1715)
Tenure^3	0.0009^{***}	0.0203^{***}	0.0006^{***}	0.0083
	(0.0002)	(0.0046)	(0.0002)	(0.0057)
Potential experience	-0.0630		0.2143	
	(0.1961)		(0.1716)	
Potential experience^2	0.0312^{***}		0.0300^{***}	
	(0.0071)		(0.0076)	
Potential experience^3	-0.0006^{**}		-0.0006^{***}	
	(0.0002)		(0.0002)	
Schooling × potential experience	0.0187		0.0101	
	(0.0177)		(0.0118)	
ASVAB × potential experience	0.0505^{***}		0.0108	
	(0.0163)		(0.0121)	
Excluded instrument F	81.60		138.85	
Observations	30,246	30,246	11,438	11,438
Respondents	7,631	7,631	2,889	2,889

Notes. Parentheses contain standard errors clustered by respondent. Sampling weights are applied. All equations include job spell, year, sex, and race indicators. Wage is real hourly wage.

Each displayed first stage has actual experience as its dependent variable. The second stage instruments (E, E^2, E^3, SE, ZE) with (P, P^2, P^3, SP, ZP), where S is schooling, Z is ASVAB, and P is potential experience. The other four first stage equations are summarized in Table 4.

The F statistic jointly tests the five excluded instruments in the displayed first stage. Blank coefficient cells indicate terms absent from that equation.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 6
IV with occupation indicators: First and second stage estimates

	NLSY79		NLSY97	
Variable Dependent variable	First stage Experience	Second stage $100\ln(wage)$	First stage Experience	Second stage $100\ln(wage)$
Schooling	0.0060	5.7046***	−0.0638	3.5700**
	(0.0612)	(0.7842)	(0.0605)	(1.5317)
Schooling × experience		0.1332		0.0201
		(0.2645)		(0.2964)
Schooling × tenure	−0.0196	−0.4033	−0.0283***	0.2638
	(0.0184)	(0.2653)	(0.0100)	(0.2579)
ASVAB	−0.0842	−1.6781***	0.0183	−0.0871
	(0.0675)	(0.5997)	(0.0534)	(1.2509)
ASVAB × experience		0.1689		0.5093*
		(0.2371)		(0.2799)
ASVAB × tenure	−0.0353	0.3329	0.0111	−0.3280
	(0.0218)	(0.2330)	(0.0106)	(0.3035)
Experience		2.4591		2.2417
		(3.6406)		(4.9116)
Experience^2		−0.2680		−0.0614
		(0.1943)		(0.2779)
Experience^3		0.0078		0.0024
		(0.0052)		(0.0069)
Tenure	0.4614**	14.3303***	0.4474***	0.0087
	(0.2145)	(3.3804)	(0.1557)	(3.9204)
Tenure^2	0.0008	−0.7044***	0.0072	−0.1210
	(0.0073)	(0.1345)	(0.0063)	(0.1938)
Tenure^3	0.0000	0.0151***	−0.0001	−0.0003
	(0.0002)	(0.0044)	(0.0002)	(0.0061)
Potential experience	0.4283*		0.3645**	
	(0.2281)		(0.1652)	
Potential experience^2	0.0090		0.0134	
	(0.0108)		(0.0094)	
Potential experience^3	−0.0001		−0.0003	
	(0.0003)		(0.0002)	
Schooling × potential experience	0.0162		0.0242**	
	(0.0202)		(0.0096)	
ASVAB × potential experience	0.0398		−0.0122	
	(0.0252)		(0.0112)	
Occupation indicators (*reference: clerical*)				
Professional		39.4476***		22.9889***
		(5.0951)		(6.8471)
Managers		25.8688***		28.9227***
		(6.3194)		(9.3032)
Sales		−8.0731		12.5663
		(7.4871)		(9.4199)
Craftsmen		31.2394***		20.9905**
		(5.3691)		(8.5769)
Operatives and laborers		7.6687**		4.8204
		(3.5814)		(6.1556)
Service		−18.8099***		−2.3662
		(3.8917)		(5.3830)
Occupation history instruments	Yes		Yes	
Observations	15,089	15,089	4,555	4,555
Respondents	3,746	3,746	1,111	1,111

Notes. Parentheses contain standard errors clustered by respondent. Sampling weights and job spell, year, sex, and race controls are included. Occupation effects are relative to clerical; slopes are common.

The first stage shown predicts experience. All 11 first stages use the full instrument set (Table 4). History coefficients are suppressed. Blank cells indicate absent terms. KP denotes Kleibergen and Paap.

For NLSY97, Stata reports a moment covariance matrix that is not full rank; the Hansen test is unavailable and inference requires caution. Relevance does not establish instrument validity.

$^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6 Appendix

6.1 Sample Journey Table

Table 7
Sample Construction

Sequential restriction	NLSY79	NLSY97
Initial sample Number	224,625	161,638
Sample weight/ASVAB/Sex/Age	215,178	106,209
Race	213,473	106,209
Hours	212,238	105,458
Wage	201,430	100,051
Occupation	196,088	68,028
Education	192,481	68,028
Tenure/Experience	192,481	67,371
Job number ≤ 3 constraint	47,002	11,440

NOTE: The table is not measured in distinct respondents. Initial counts refer to the merged panels after earlier cleaning and employer selection. The final row retains existing job numbers 1 through 3

References

Altonji, Joseph G. and Charles R. Pierret (2001) "Employer Learning and Statistical Discrimination," *The Quarterly Journal of Economics*, 116 (1), 313–350, 10.1162/003355301556329.

Farber, Henry S. and Robert Gibbons (1996) "Learning and Wage Dynamics," *The Quarterly Journal of Economics*, 111 (4), 1007–1047, 10.2307/2946706.

Gibbons, Robert, Lawrence F. Katz, Thomas Lemieux, and Daniel Parent (2005) "Comparative Advantage, Learning, and Sectoral Wage Determination," *Journal of Labor Economics*, 23 (4), 681–724, 10.1086/491606.

Spence, Michael (1973) "Job Market Signaling," *The Quarterly Journal of Economics*, 87 (3), 355–374, 10.2307/1882010.

Zhang, Ye (2009) "Employer Learning Under Asymmetric Information: The Role of Job Mobility," *SSRN Electronic Journal*, 10.2139/ssrn.1058801.